14.01 — Principles of Microeconomics

90 points

Midterm Exam — Graded

Fall 2023

Student: Amin Cheipournejad · Total: 67 / 90

Grade Report

Part	Score	Grader's comment
Q1 1.1	**5 / 5**	Correct: with min{x, y} the goods are perfect complements, so the consumer always buys them in equal amounts, $x = y = m/(p_x + p_y)$, however the prices compare, and never spends everything on x.
Q1 1.2	**5 / 5**	Correct: q = L is constant returns to scale, so profit is zero and no firm wants to enter. Zero profit also means no firm wants to leave, which the answer leaves implicit; the number of firms stays at J.
Question 1 subtotal	**10 / 10**	

Part	Score	Grader's comment
Q2 2.1	**3 / 3**	Correct: $p_y y = m$ while food spending stays within the stamps, and $p_x x + p_y y = m + f$ beyond them. The graph has the right shape, flat and then falling with slope $-p_x/p_y$, with the kink marked. Its points are not labelled (the kink at f/p_x, the height m/p_y, the end at $(m + f)/p_x$), but the equations above it fix them.
Q2 2.2	**5 / 5**	Correct: the tangency $3y/x = p_x/p_y$ gives $x = 3(m + f)/(4p_x)$ and $y = (m + f)/(4p_y)$, and the answer makes the check that matters: food spending, $3(m + f)/4$, exceeds f exactly when $3m > f$, so the stamps never bind.
Q2 2.3	**5 / 5**	Correct: $m = (4/3)x - f$, so each rise in f shifts the Engel curve down in parallel, and the intuition is the official one: more stamps mean less of his own income is needed for any given amount of food.
Q2 2.4	**9 / 10**	The numbers are all right: (900, 300) before, (450, 300) after, and the compensated bundle (756.8, 504.5) on the original indifference curve at utility 683.85, giving a substitution effect of −143.2 and an income effect of −306.8 on food. The subtractions are written the other way round from the negative results they report; read as changes, the signs are right. The graph falls short of the decomposition: it shows the three bundles but not the compensated budget line, the line at the new prices that touches the original indifference curve at (756.8, 504.5), which is what separates the move along the curve from the parallel shift. It also starts the new budget line lower on the ticket axis, where it should pivot about the same point since only food got dearer, and puts (450, 300) below (900, 300) though tickets do not change. One point.
Q2 2.5	**1 / 5**	The condition is wrong. The answer holds food at 900, solving $3(1000 + f)/8 = 900$ for $f = 1400$; the question holds utility at its old level, 683.85. Holding food fixed overshoots: at f = 1400 Pedro buys 900 of food and 600 tickets, a utility of about 813. The compensated bundle from part 4 costs about 2018.2 at the new prices ($2 \times 756.81 + 504.54$), so the stamps must rise by about 818.2, to about 1018.2. The answer also gives the new level of stamps rather than the increase. The demand for food at the new price, as a function of f, is right. Four points.
Q2 2.6	**2 / 2**	Correct: the budget becomes $p_x x + p_y y = m + s$, so $x = 3(m + s)/(4p_x)$ and $y = (m + s)/(4p_y)$.
Q2 2.7	**5 / 5**	Correct: $s = f$, for the right reason: Pedro already spends more on food than the stamps are worth, $3(m + f)/4 > f$ because $3m > f$, so the restriction never binds and the stamps are as good as cash.
Question 2 subtotal	**30 / 35**	

Part	Score	Grader's comment
Q3 3.1	**3 / 3**	Correct: with capital fixed at K, solving log(1 + L) + log(1 + K) = q gives L = e^q/(1 + K) − 1, so the short-run cost is w(e^q/(1 + K) − 1) + rK, which is w(e^q − 1) with no capital; keeping K general is valid. The line before it writes 1 + L where it means L, and the cost line is read with the −1 outside the exponent, as it is in that line.
Q3 3.2	**0.5 / 5**	Wrong. With no capital the cost is w(e^q − 1), but the working uses w·e^{q-1}, with the −1 moved into the exponent, and then differentiates e^{q-1} by the power rule, giving (q − 1)w·e^{q-2}. The derivative of e^q is e^q itself: marginal cost is w·e^q and average cost is w(e^q − 1)/q. The output question is not answered: a firm that produces sets price equal to marginal cost, p = w·e^q, so q = log(p/w). The condition given, producing when marginal cost exceeds average cost, is not that rule. Average cost as cost over quantity is the right idea, applied to the wrong cost. Four and a half points.
Q3 3.3	**2 / 7**	The start is right: setting the marginal rate of technical substitution equal to w/r gives 1 + K = (w/r)(1 + L), and substituting it into the production function is the right next step. The answer stops there, so neither thing the question asks for is reached. With w = r = 1, $(1 + L)^2$ = e^q, so L = K = $e^{q/2}$ − 1 and the long-run cost is C(q) = 2($e^{q/2}$ − 1). Marginal cost is $e^{q/2}$ and average cost rises from 1, so the long-run supply curve is q = 2 log p for p ≥ 1, and zero below. Five points.
Question 3 subtotal	**5.5 / 15**	

Part	Score	Grader's comment
Q4 4.1	**1 / 3**	Marginal cost, 2q, and the rising part of supply, p = 2q, are right. The rest is not. The answer says yes, the firm produces at any price, but because it pays the 9 whenever it produces anything, price must cover average cost, q + 9/q, which is lowest at q = 3, where it equals 6. Supply is q = p/2 for p ≥ 6 and zero below, so the answer is no. Part 2 uses exactly that threshold. Two points.
Q4 4.2	**2 / 2**	Correct: four firms each supplying p/2 give Q = 2p for p ≥ 6, and zero below.
Q4 4.3	**0 / 5**	Wrong. Setting demand equal to the aggregate supply from part 2, 30 − p = 2p, gives p = 10 and Q = 20. The answer's p = 15 and Q = 120 fit neither curve: at p = 15 demand is 15 and supply is 30. Each firm produces 5 and makes 10 × 5 − (5^2 + 9) = 16. The profit line has a second error: the cost of 129 does not come from C(q) = q^2 + 9; at the answer's own 30 units per firm it would be 909. Five points.
Q4 4.4	**5 / 5**	Correct: revenue q(30 + A − q) gives marginal revenue 30 + A − 2q; setting it equal to marginal cost, 2q, gives q = (30 + A)/4 and p = (90 + 3A)/4. The working labels the condition loosely, but the derivative taken and the results are right.
Q4 4.5	**3.5 / 5**	Consumer surplus, $(30 + A)^2$/32, and producer surplus, $(30 + A)^2$/8, are right, and so is the answer that the outcome is not efficient, with the lost triangle sketched. The deadweight loss the question asks for is never computed. The triangle runs from the monopoly quantity, (30 + A)/4, where price exceeds marginal cost by (30 + A)/4, to the efficient quantity, (30 + A)/3, where demand meets marginal cost, so the loss is ½ × (30 + A)/12 × (30 + A)/4 = $(30 + A)^2$/96. One and a half points.
Q4 4.6	**7 / 7**	Correct: competitive consumer surplus is 200, the monopoly's is $(30 + A)^2$/32, and the second is larger exactly when A > 50. The intuition is the official one: the extra value consumers place on lactose-free ice cream has to outweigh the output the monopolist holds back by setting marginal revenue equal to marginal cost. The 200 is the surplus at the correct equilibrium, p = 10 and Q = 20, not at the one written in part 3.
Q4 4.7	**3 / 3**	Correct: the cap goes where demand meets marginal cost, p = (60 + 2A)/3, and the intuition is right: at that cap, holding output back no longer raises the price, so the monopolist produces the efficient quantity and the deadweight loss disappears.
Question 4 subtotal	**21.5 / 30**	

Report Card

Section	Earned	Available	Percent
Question 1 — True or False	**10**	**10**	**100%**
Question 2 — Consumer Theory	**30**	**35**	**86%**
Question 3 — Short-Run Producer Theory	**5.5**	**15**	**37%**
Question 4 — Long-Run Production Theory	**21.5**	**30**	**72%**
Total	**67**	**90**	**74%**

Each part was assessed independently against the official solution set and scored on its own merits, without reference to the running total. Full marks were awarded wherever the work is correct and complete, and where the student's method departs from the official solution but is valid it earns full credit, as with the short-run cost in 3.1, which keeps the fixed capital general. Deductions were taken only for an error on the page or for something a question asks for that the answer does not supply; every deduction is itemised above with the specific gap named, and an error carried into a later part was penalised only where it first appears. The official solution stops short in two places, giving 2.5 as a formula without the number and 3.3 without the supply curve, so those two answers were derived independently; its compensated ticket figure of 504.6 is a rounding slip for 504.54, so the submission's 504.5 was taken as correct. Where the handwriting was ambiguous, as with the position of the −1 in the cost expressions of 3.1 and 3.2, the reading consistent with the student's surrounding work was taken. The handwritten submission was read from rendered page images, and every derivation was re-checked independently before scoring, including the compensated bundle and the food-stamp increase in Question 2 and the long-run cost and supply curve in 3.3, which were checked numerically. I consider this a fair and defensible assessment of the work as submitted.

Claude (Opus 5.5) — Anthropic
Grader, at the student's request
Graded 22 October 2026

MIT 14.01, Fall 2023 — Midterm Exam. Assessed against the official solution set. Marks read from the student's handwritten submission; no content added, removed, or amended. A self-study assessment by an AI grader — not an official grade, and carries no standing with MIT. Source: MIT OpenCourseWare, 14.01 Principles of Microeconomics, Fall 2023.